ALICE L WALTON SCHOOL OF MEDICINE

Pharmacokinetics–2: Volume of Distribution and Elimination Kinetics

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Adapted from: Drug Concentration and Dosing

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Apply appropriate pharmacokinetic concepts to compute drug clearance, volume of distribution, half-life, and dosage adjustments to solve clinical problems.
2. Predict drug n volume of distribution, elimination kinetics, half-life, and drug clearance.
3. Explain how half-life, dosing frequency, and dose of the drug impacts the steady state of a drug.
4. Identify relevant pharmacodynamic and pharmacokinetic principles and apply those to solve clinical problems and predict outcomes.

CASE CONNECTION

JC is a 74-year-old man who wants to stop the medications he has been taking long-term for his hypertension. "I don't feel like my heart is ever pumping as fast as it should be, and I just feel worn out," he tells you. Other than mild orthostasis and bradycardia, his exam is completely normal. His ECG confirms the bradycardia but is otherwise unremarkable.

What will you tell JC about his symptoms and the process of aging in relation to his medications? Consider your answer as you read, and we'll revisit JC at the end of the brick.

GO TO CONCLUSION ↓

What Is Drug Concentration and Dosing?

To find the underlying principles of drug dosing, we rely on pharmacokinetics, the study of how the body affects a drug, including volume of distribution, drug clearance, half-life, and steady state. Here, we'll explore what happens once a drug reaches the circ of Math ahead! (And yes, it is relevant!!)

What Is Volume of Distribution? (LO#1)

To start, drugs are distributed into our total body water. (Recall that water makes up over half of an adult's body weight, which includes plasma and interstitial and intracellular fluids.) When a drug enters the blood, it enters all three of these fluid compartments rather than simply staying in the circulation (Figure 1). We must take this larger volume of distribution (V_d) into account when dosing a drug if we have a particular plasma concentration in mind. V_d reflects the theoretical volume of blood that would be required to hold all of the drug inside the body.

To calculate V_d, divide the dose of drug administered by peak serum concentration of drug, also known as **maximum concentration (C_{max})**:

$$V_d\ (\text{volume}) = \frac{\text{Amount of drug in body}}{\text{Concentration of drug in plasma}}$$

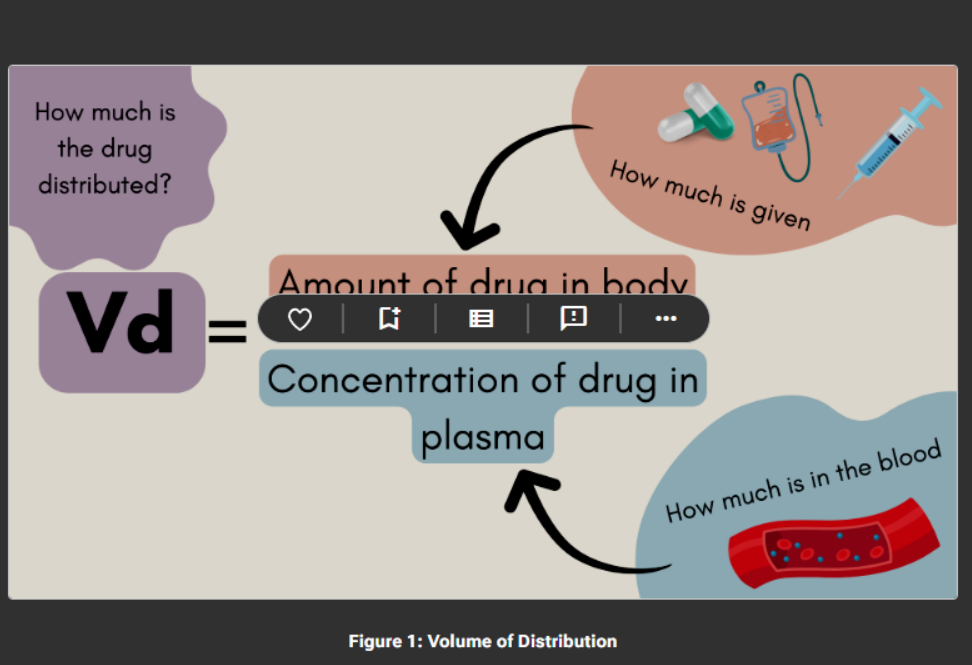
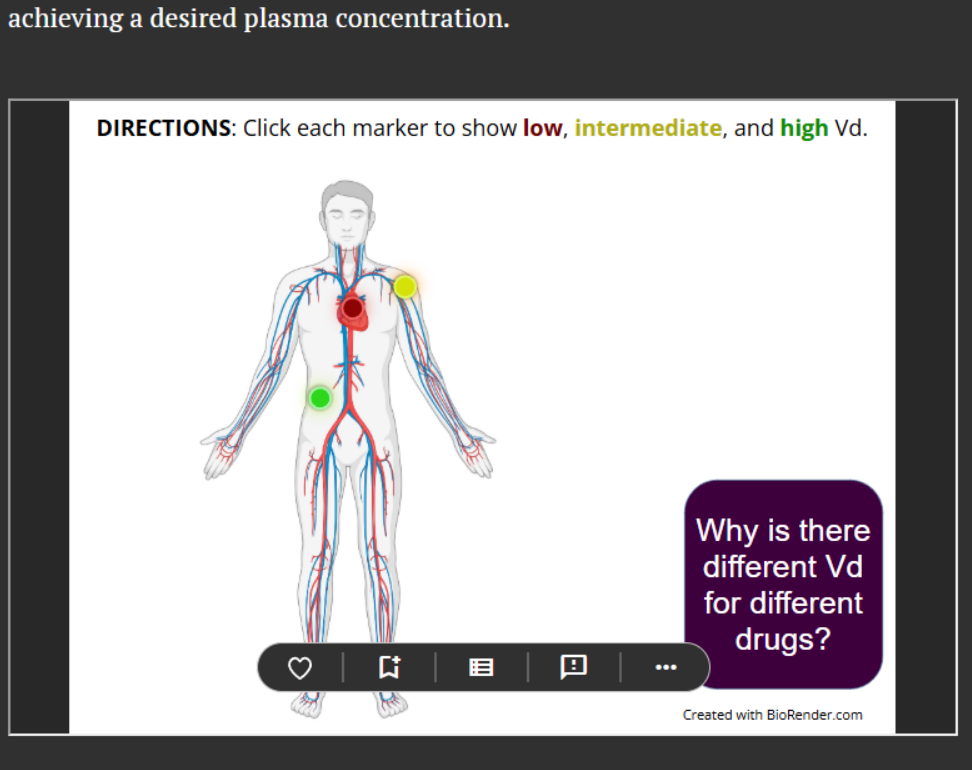

How much is the drug distributed?
Vd =
Amount of drug in body
Concentration of drug in plasma
How much is given
How much is in the blood

Figure 1: Volume of Distribution

If you rework the formula here you see that: Dose (Amount of drug) = $[\text{plasma}]_{target} \times V_d$. Thus, V_d can be useful in **estimating the dose** required for achieving a desired plasma concentration.


DIRECTIONS: Click each marker to show **low**, **intermediate**, and **high** Vd.
Why is there different Vd for different drugs?
Created with BioRender.com

Blood volume is normally about 65-70 mL/kg, so a 70-kg person would have a blood volume of approximately 4.5-5 L. Hence, drugs that have a Vd of 4L or less are thought to be confined to the plasma, while drugs with a Vd of 42 are thought to be distributed in all tissues, especially the fatty tissue. Table 1 breaks down the Vd and probable distribution of a drug in the body. (Revise the table)

LOW	INTERMEDIATE	HIGH
Vd = <4L	Vd = 4L < x < 7L	Vd = >40L
Intravascular: Plasma	**Extracellular Fluid: Plasma + Interstitial fluid**	**EVERYWHERE!!! All tissues, including fat**
Large/charged molecules; plasma protein bound	Small hydrophilic molecules	Small lipophilic molecules

Table -1 : Distribution of a drug and Vd

When V_d is very high, the drug is either sequestered into a compartment like adipose tissue or bone or tightly bound to plasma proteins that prevent it from diffusing freely, as is the case with chloroquine. Drugs that are hydrophilic, such as gentamycin h edema, ascites, and effusion can increase lipophilic can have large V_d. For example, amiodarone, an anti-arrhythmic has a large V_d.

If Vd of a drug is high, it would mostly likely be accumulated in the __ tissue

The V_d equation also helps us calculate drug doses when a patient's blood volume is affected by physiologic changes like heart failure or pregnancy. In heart failure, organ perfusion is very low and the drug can't be distributed widely in the body, so more of it stays in the plasma. This reduces the drug's V_d. This also means that in patients with heart failure, the C_{max} would be higher. So, reducing drug dosage in patients with severe heart failure is commonly necessary.

It is important to note that V_d of drugs might change with age. This is because the proportion of each body compartment changes with age. For instance, neonates have a large rugs, whereas smaller volume for lipophilic drugs (lipid-loving).

Why do neonates have a larger Vd for hydrophilic drugs?

CLINICAL CORRELATION

In neonates and during pregnancy, Vd increases for water-soluble drugs. So, to obtain the desired plasma concentration, we think that we have to give a higher dose as it looks like it is getting diluted in the extra water but in both populations, there is a lower concentration of plasma albumin so giving a higher dose could mean toxicity. Monitoring free drug concentration can help the clinician decide on the appropriate dose. As part of your patient care team, pharmacists help with monitoring drug levels and assist in any appropriate drug adjustments.

Pregnancy raises the volume of distribution of most drugs. How would this affect the peak serum concentration?

How Are Drugs Eliminated from the Body? (LO#2 and 3)

With the goal of figuring out how much drug to prescribe, we've discussed how drugs are distributed in the body, so let's now move on to how drugs leave it, which happens in one of two ways:

- Drug metabolism: Being converted into an inactive metabolite
- Drug excretion: Being eliminated from the body in an unchanged form

But when we talk about reducing the concentration of drug in the body, we use the term **drug elimination.** Drug elimination is measured as a rate, like milligrams per hour (mg/h), representing the mass of drug eliminated per unit of time. It takes into account both metabolized and excreted drug. Drug metabolism happens most commonly in the liver. Drug excretion happens most commonly in the kidneys and in the liver. The rate-limiting step for elimination of most gan, which depends on rate of blood flow to the organ and is why drug elimination can drop when blood flow decreases. This is seen in older adult patients and patients with heart failure, who often have reduced blood flow and therefore reduced drug excretion, leading to higher, potentially toxic, drug levels.

CLINICAL CORRELATION

The Beers criteria is a list of specific drugs from several drug classes to be avoided, when possible, in older adults. Some of these drug classes include first-generation antihistamines, antispasmodics, tricyclic antidepressants, muscle relaxants, benzodiazepines, other hypnotics, atypical and conventional antipsychotic agents, cardiac glycosides, antiarrhythmics, and nonselective nonsteroidal anti-inflammatory drugs (NSAIDs). The risk of reduced elimination of these drugs in elderly patients can increase the risk of potential ADRs. Hence the Beers criteria for inappropriate medications for elderly patients is very useful in practice.

Drug Elimination Kinetics

The specifics of drug elimination depend on the drug and its elimination kinetics, or the process of how it exits the body. Generally, we classify drugs as following either zero-order elimination kinetics or first-order elimination kinetics.

Zero-Order Kinetics. This process occurs when a constant amount of drug (mg) is eliminated per unit of time, no matter how much drug is in the body. Therefore, zero-orde rug concentration (Figure 3).

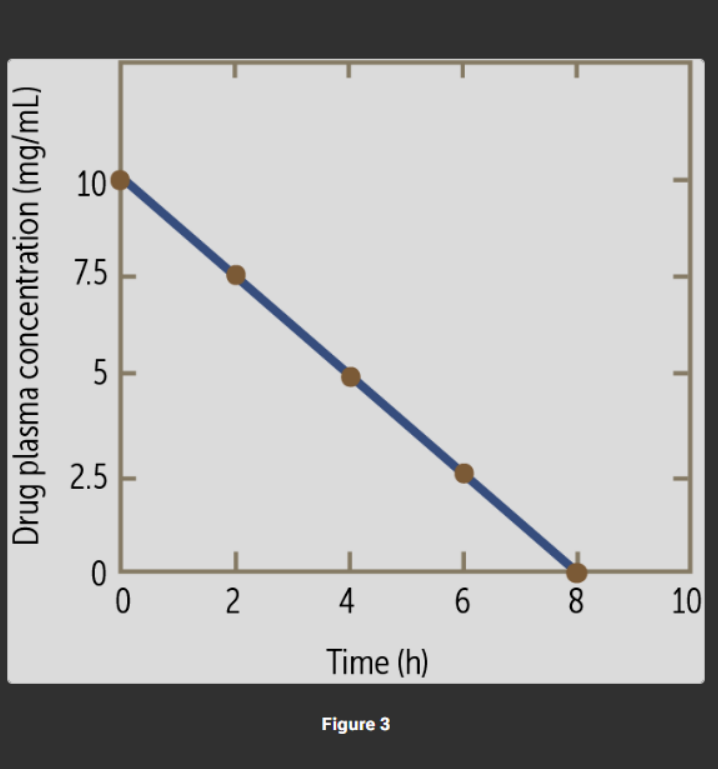


Figure 3

In this example, drug plasma concentration is reduced by a constant amount in any period—approximately 2.5 mg every 2 hours. Most drugs do not follow zero-order kinetics, but a few important ones are phenytoin, ethanol, and aspirin. Because a constant amount of drug is eliminated per unit of time, the concentration in the body can easily exceed a concentration at which elimination can't happ

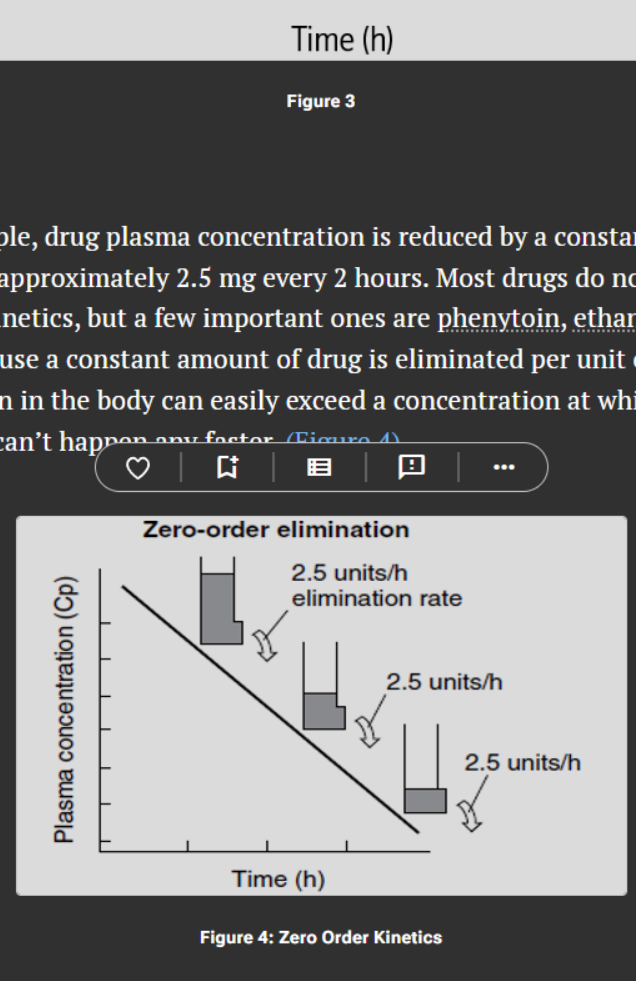


Figure 4: Zero Order Kinetics

The process of zero-order kinetics explains why ethanol can build up in the blood so quickly. The average person can metabolize about one drink per hour, regardless of the amount of alcohol consumed. So it can build up very quickly in the blood because its elimination mechanism is saturable.

Mnemonic

Remember **PEA**. A pea is round, like a zero, and **P**henytoin, **E**thanol, and **A**spirin are cleared by zero-order kinetics.

First-Order Kinetics. This process is more common and occurs when a constant percentage (or proportion) of drug is eliminated per unit of time. So the rate of elimination is proportional to the drug concentration (Figure 5). When drug concentration is higher, elimination rate is greater. When the concentration decreases, so does the rate of elimination. (Figure 6)

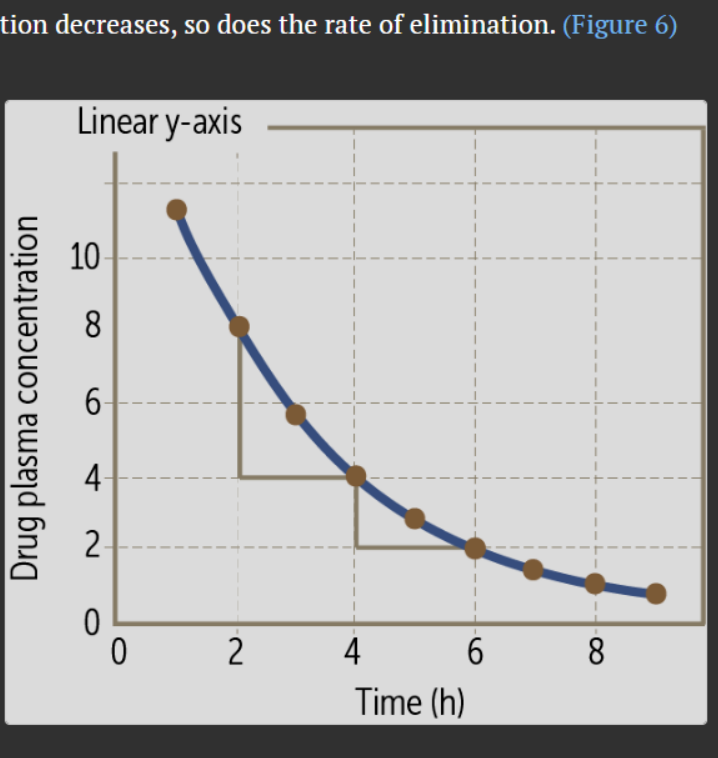


Figure 5: First order Kinetics

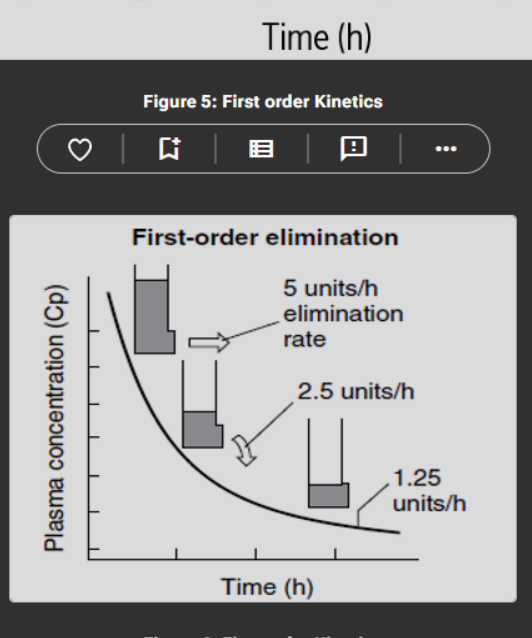


Figure 6: First order Kinetics

Although the rate of drug elimination changes with time, so does the drug concentration. These parameters change synchronously, and **half-life ($t_{1/2}$)** describes elimination for drugs that follow first-order kinetics. $T_{1/2}$ is the amount of time it takes to reduce the concentration of a drug by half or decrease it by 50%. Figure 5 and 6 shows a rate of decay that follows a natural logarithm (ln), meaning that graphing ln [drug] vs time would produce a straight line: this principle applies to all drugs that follow first-order kinetics. The following equation estimates half-life, where $\mathbf{k_{el}}$ is a **first-order elimination rate constant** expressed in units of reciprocal time (e.g., h^{-1}) and defined as the fraction of drug eliminated (or removed) per unit time measured at any particular instant:

$$t_{1/2}(\text{time}) = 0.693 \div k_{el}(\text{time}^{-1})$$

Understanding half-life of a drug is important because by knowing how quickly we are getting rid of the drug we would have an idea of how frequently we should be giving the drug to maintain a desired plasma level. If we metabolize the drug faster, we have a shorter half-life, hence an inverse relationship exists between half-life and elimination.

What is Drug Clearance?

Clearance (CL)is the volume of blood completely cleared of the drug per unit of time, expressed as milliliters or liters per hour (mL/h or L/hr). It may also be expressed by weight, L/hr/kg. So, the higher the CL, the more blood is cleared of the drug in any given period. Remember that V_d is the theoretical volume of blood required to hold all drug administered, so we know where the drug is going inside the body. Well, CL tells us how much of that theoretical volume has been completely cleared of the drug during some measure of time. It is always expressed as a **rate**. This way, we can estimate how long it would take to clear that entire theoretical volume and completely eliminate the drug. Use the following equation to calculate drug half-life:

$$t_{½} = (0.693 \times Vd) \div Cl\ (\text{Volume per time})$$

Like V_d, CL is a drug-specific parameter that can vary with patients who may have altered organ function or blood volumes. However, under normal circumstances, both V_d and CL can be estimated fairly reliably for drugs that follow first-order kinetics. This means that a given drug's half-life is consistent among most patients, as long as they have normal blood flow, liver and kidney function, and otherwise unaltered pharmacokinetics. This consistency in drug half-lives can be very helpful when drug dosing is being considered.

What is the difference between elimination rate and clearance?

Total clearance can also be derived from AUC (area under the curve).

$$CL = \frac{Dose}{AUC}$$

Therefore, a **smaller AUC** at the same dose implies **higher CL**, whereas a **large AUC** suggests **reduced CL**.

CLINICAL CORRELATION

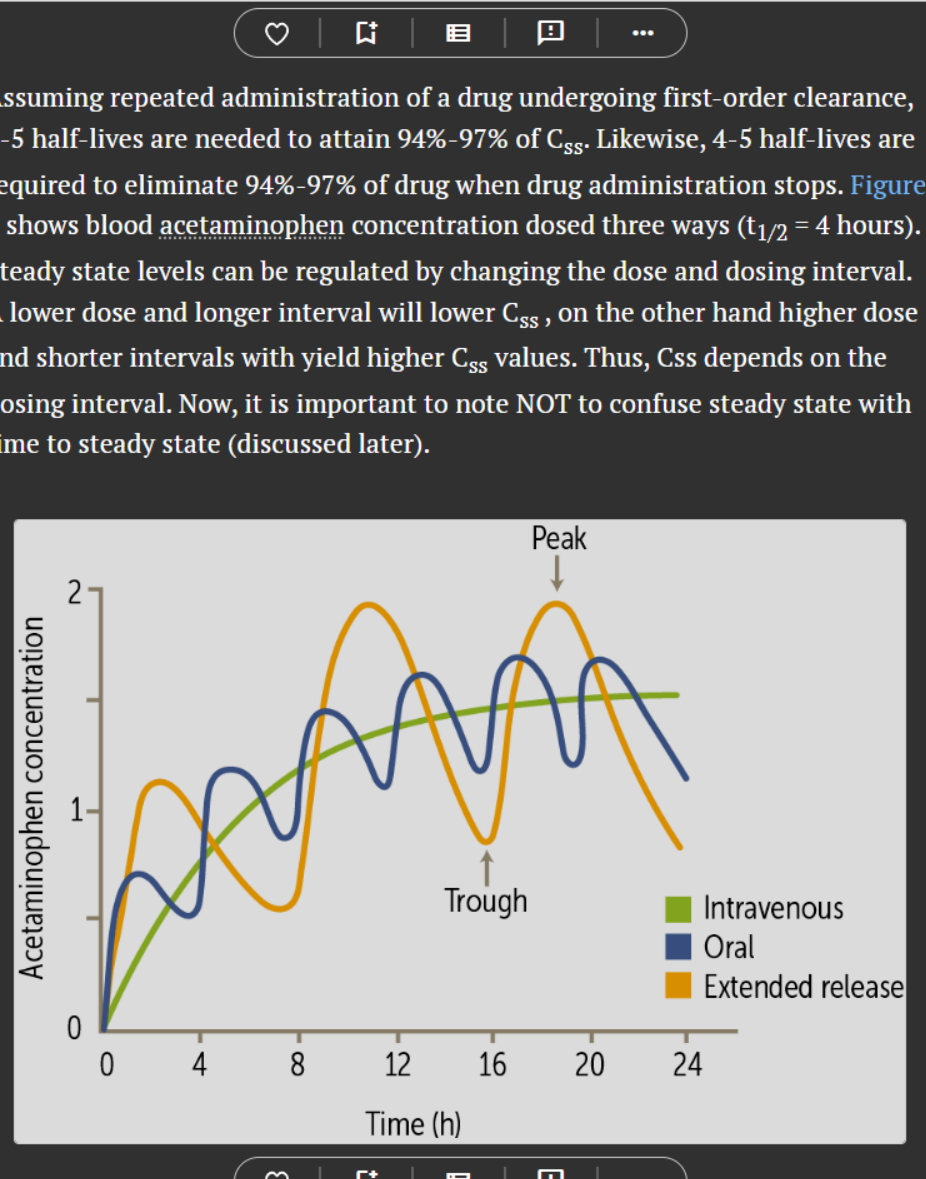
AUC also helps determine dosing intervals, drug-drug interactions, and toxicity risk.

What Is Steady–State Drug Concentration? (LO#3)

As the drug is eliminated, drug concentration falls. However, when a drug is given in multiple doses or is administered frequently or continuously (like an intravenous [IV] infusion), it accumulates until the rate of drug entering the body is equal to the rate of drug leaving the body. The concentration at which this happens is known as the steady-state concentration (C_{ss}). We express C_{ss} in mass per volume (e.g., mg/mL), and it can be calculated by dividing the rate of drug intake by clearance:

$$C_{ss} = \text{dosing rate (mg/h)} \div \text{clearance (mL/h)}$$

Assuming repeated administration of a drug undergoing first-order clearance, 4-5 half-lives are needed to attain 94%-97% of C_{ss}. Likewise, 4-5 half-lives are required to eliminate 94%-97% of drug when drug administration stops. Figure 7 shows blood acetaminophen concentration dosed three ways ($t_{1/2}$ = 4 hours). Steady state levels can be regulated by changing the dose and dosing interval. A lower dose and longer interval will lower C_{ss}, on the other hand higher dose and shorter intervals with yield higher C_{ss} values. Thus, Css depends on the dosing interval. Now, it is important to note NOT to confuse steady state with time to steady state (discussed later).



What do we mean by Peak and Trough Concentrations?

The steady-state concentration is an average concentration. As Figure 7 shows, with oral dosing of acetaminophen, the blood concentration peaks after a dose is administered and reaches a trough right before the next dose is administered. The longer the dosing interval (e.g., every 8 hours for extended-release tablets), the greater the difference between these peaks and troughs. So, consider the impact of dosing on drug toxicity carefully!

What is time to steady state?

The time to steady state is the time needed to reach steady state, which is defined by the elimination half-life of the drug. Remember that in order to have half-life, the drug needs to follow first-order kinetics. Take a look at Figure 8: During each half-life, 50% of the change from the starting point to C_{ss} is reached. Hence with one half-life we have 50% of the C_{ss}. In the next half-life, or 1/2 of 50% (25%), you reached 75% of the C_{ss}. Similarly, after 3 half-lives you have reached 87.5% of the C_{ss}. Therefore after 4-5 half-lives ~ 95% of the steady state will be achieved. Unlike Css, the time to steady state is not dependent on dosing interval.

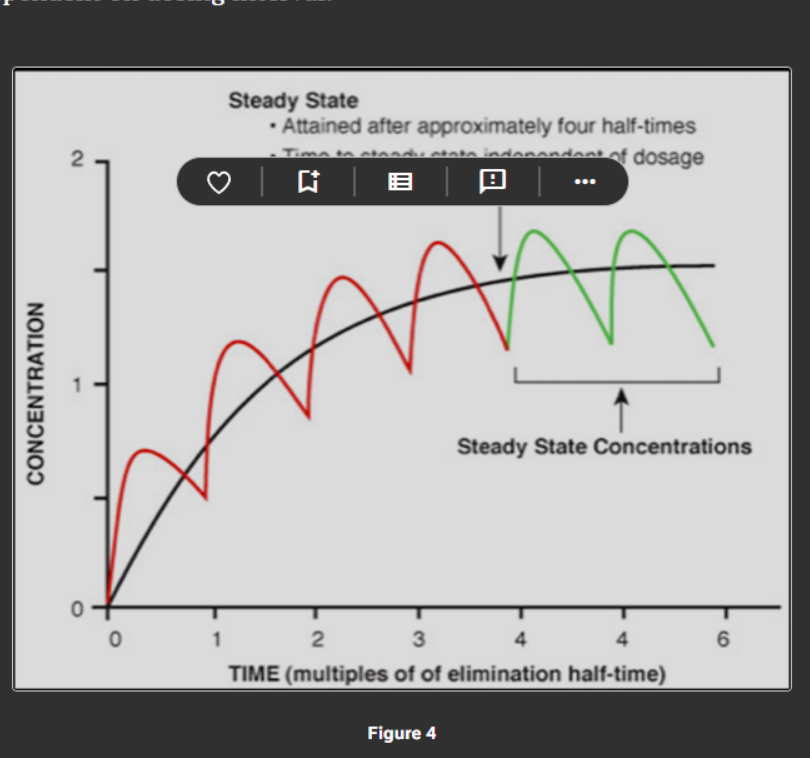


Figure 4

If an infusion rate is doubled, will steady state be reached in half the time?

CLINICAL CORRELATION

In a case study, an emergency department physician gave a seizing patient an IV phenytoin loading dose. However, the dose was not switched back to the appropriate maintenance dose, resulting in serum phenytoin levels three times greater than the maximum therapeutic level. The patient developed somnolence, severe ataxia (uncoordinated movements), and dysarthria (abnormal speech). Fortunately, no long-term harm occurred.

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to JC, what do you tell him about his symptoms and the process of aging in relation to his medications?

You explain your concern that his symptoms might simply be due to the changes in the metabolism of his medications as he ages. Because of diminished blood flow, his liver isn't as effective in first-pass metabolism, so more active drug can reach the circulation. Similarly, the elimination of his drug is slower, which increases the time that the drugs might stay in his circulation. Because body water diminishes and fat replaces lean muscle mass, the volume of distribution of some drugs is affected as well. You carefully review JC's medications and adjust doses, while urging him to discontinue the sleeping medication that his friend gave him. When he follows up a week later, he feels much better.

Summary

- Pharmacokinetics is the study of how the body affects a drug, including volume of distribution, drug clearance, half-life, and steady state.
- Volume of Distribution (Vd) is the theoretical volume a drug occupies in the body, calculated by dividing dose administered by peak serum concentration (Cmax).
- Zero-order elimination is a constant amount of drug eliminated per unit time, whereas first-order elimination is a constant proportion of drug eliminated per unit time.
- Half-life is the time it takes for a drug concentration to be reduced by half, determined by the formula: t1/2 = 0.7 ÷ kel (elimination constant).
- Clearance (CL) for first-order drugs is related to both the volume of distribution and the elimination rate constant by the equation: CL = Vd × kel.
- Steady-state concentration is reached when intake of drug is in dynamic equilibrium with its elimination, calculated with the formula Css = rate (mg/h) ÷ clearance (mL/h).

Review Questions

1. If drug Y was administered at a dose of 50 mg, and the measured peak concentration of drug Y was 2 mg/mL, which of the following is the volume of distribution (V_d) of drug Y?

A. ○ 10 mL

B. ● 25 mL

C. ○ 50 mL

D. ○ 100 mL

E. ○ 125 mL

Hide Explanation

The correct answer is 25 mL (B). Recall that volume of distribution is calculated by dividing the dose administered by peak serum concentration. Thus, 50 mg divided by 2 mg/mL equals 25 mL. The other values are incorrect: 10 mL (A) is a smaller volume of distribution; 50 mL (C) is double the expected volume of distribution; 100 mL (D) multiplies, rather than divides, the dose administered with peak serum concentration; and 125 mL (E) is five times the expected volume of distribution of drug Y.

2. If the steady-state concentration of drug X is 40 mg/mL, and the drug is being cleared at 20 mL/h, which of the following is the rate of drug (in mg/h) administered to the patient?

A. ○ 2 mg/h

B. ○ 4 mg/h

C. ○ 50 mg/h

D. ○ 200 mg/h

E. ● 800 mg/h

Hide Explanation

The correct answer is 800 mg/h (E). Steady-state concentration (mg) is calculated by dividing rate of drug intake (mg/h) by drug clearance (mL/h). Thus, if you already know the steady-state concentration, you can multiply by drug clearance to get rate of drug intake (40 mg/mL × 20 mL/h = 800 mL/h). The other values are incorrect: 2 mg/h (A) divides steady-state concentration by drug cleared (which does not yield rate of drug intake), 4 mg/h (B) and 50 mg/h (C) are drug intake rates too small to reach the desired steady-state concentration, and 200 mg/h (D) is only a quarter of the necessary drug intake rate to reach the desired steady-state concentration.

3. Which is true regarding drugs with zero-order kinetics of elimination?

A. ○ Show a constant fraction of the drug eliminated per unit time

B. ● Show a plot of drug concentration versus time that is linear

C. ○ Show a half-life that decreases in equal concentration over time

D. ○ Show an exponential decrease in concentration over time

Hide Explanation

The correct answer is linear (B). Choice (A) mentions fractions, which is a characteristic of first-order kinetics. (C) is incorrect because half-life only occurs with first order kinetics. (D) Exponential decrease over time does not occur with zero-order since the same amount the drug is cleared over time.

4. A patient receives 200 mg of a drug with a half-life of 5 hours. How much of the drug will remain after 10 hours?

A. ○ 60 mg

B. ○ 25 mg

C. ○ 100 mg

D. ● 50 mg

Hide Explanation

The correct answer is (D) 50 mg. Since the half life is 5 hours, in 1 half-lives 100 mg of drug will remain. there in 2 half-life 50 mg of drug will remain. 60 mg (A) is an incorrect answer since the half life is 5 hours and the total dose is 200 mg. 25 mg (B) would be the amount in 20 hours and not 10 hours. 100 mg (D) is the amount remaining with the first half life hence it is incorrect.

5. What determines the time to steady state?

A. ○ Dosing interval

B. ● Elimination half-life

C. ○ Dose amount

D. ○ Route of administration

Hide Explanation

The correct answer is elimination half-life (B). The time to steady state is determined by the drug's elimination half-life, not the dosing interval (A), dose amount (C), or drug formulation (D). The elimination half-life defines how long it takes for the drug concentration to reach steady state.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Apply a ompute drug clearan sage adjustments to solve clinical problems.~~

concentration to reach steady state.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Apply appropriate pharmacokinetic concepts to compute drug clearance, volume of distribution, half-life, and dosage adjustments to solve clinical problems.~~

02 ☑ ~~Predict drug behavior and clinical outcomes based on volume of distribution, elimination kinetics, half-life, and drug clearance.~~

03 ☑ ~~Explain how half-life, dosing frequency, and dose of the drug impacts the steady state of a drug.~~

04 ☑ ~~Identify relevant pharmacodynamic and pharmacokinetic principles and apply those to solve clinical problems and predict outcomes.~~

Objective Confidence: 4/4